Monte Carlo Neutron Transport

Machine Learning Surrogate Model

Project Report

10,000	6	3	98.8%
Simulations Run	Input Features	Models Tested	Best Accuracy

A plain-English guide to the simulation, the equations, the data, and the machine learning results.

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1. What Is This Project?

Core idea: We simulate how neutrons travel through nuclear materials using random probability — then train a machine learning model to predict the result instantly, without running the full simulation every time.

This project combines **nuclear physics simulation** with **machine learning** to predict the *fission rate* — the fraction of neutrons that split atomic nuclei — from material properties alone.

Think of it like this: imagine you need to know how a ball will bounce inside a pinball machine with thousands of different configurations. Running the full machine every time is slow. If you have done it 10,000 times already, you can train a model to *predict* the result instantly from the setup alone. That is exactly what we did — but with neutrons instead of pinballs.

What is the fission rate?

The fission rate is the fraction of simulated neutrons that cause a nucleus to split. It ranges from 0.0 (no fission at all) to around 0.53 (more than half of neutrons cause fission). It is the key output of a reactor physics simulation and the **target variable** our model learns to predict.

2. The Monte Carlo Method — Plain English

Monte Carlo is a technique that uses **random numbers to simulate real-world events**. It was named after the famous Monte Carlo casino in Monaco, because both involve probability and chance.

In this project, we simulate thousands of individual neutrons. Each neutron travels through a material, bouncing around randomly until one of three things happens:

Event	What Happens	Outcome
Scatter	Neutron bounces off a nucleus like a billiard ball	Keeps moving
Absorption	Nucleus swallows the neutron	Neutron disappears
Fission	Nucleus splits apart, releasing energy	Chain reaction

3. Equations Used

3.1 Cross-Sections — Measuring How a Material Interacts

Every material is described by *cross-sections* — numbers that measure how strongly it interacts with neutrons. The higher the cross-section, the more likely the interaction. Units are cm^{-1} (interactions per centimetre of material).

Symbol	Name	Meaning
Σ_s	Scattering	How strongly the material bounces neutrons
Σ_a	Absorption	How strongly the material swallows neutrons
Σ_f	Fission	How strongly the material causes nuclear splitting
Σ_t	Total	All interactions combined

The total cross-section is simply the sum:

$$\Sigma_t = \Sigma_s + \Sigma_a + \Sigma_f \tag{1}$$

3.2 Distance to Next Collision — The Core Formula

This is the heart of the Monte Carlo method. For each neutron, we sample how far it travels before hitting a nucleus using the **exponential distribution**:

$$s = -\frac{\ln(\xi)}{\Sigma_t} \tag{2}$$

where $\xi \sim \mathcal{U}(0, 1)$ is a uniform random number drawn fresh for each step, and s is the sampled distance in centimetres.

In plain English: pick a random number between 0 and 1, take its natural logarithm, flip the sign, and divide by how reactive the material is. A denser material (high Σ_t) means shorter distances — the neutron hits something sooner.

The negative logarithm transforms a uniform random variable into an exponentially distributed one, which is the correct physical distribution for particle free paths under the assumption of spatially uniform material.

3.3 Collision Outcome Probabilities

After landing, a second uniform random number η decides what happens at the collision. Each interaction type gets a probability proportional to its cross-section:

$$P(\text{scatter}) = \frac{\Sigma_s}{\Sigma_t}, \qquad P(\text{absorb}) = \frac{\Sigma_a}{\Sigma_t}, \qquad P(\text{fission}) = \frac{\Sigma_f}{\Sigma_t} \tag{3}$$

These three always sum to exactly 1.0 — every neutron must have one outcome. In code:

- If $\eta < P(\text{scatter}) \rightarrow$ scatter, sample a new distance
- Else if $\eta < P(\text{scatter}) + P(\text{absorb}) \rightarrow$ absorbed, neutron ends
- Else $\rightarrow$ fission event, neutron ends

3.4 Theoretical Fission Rate — Infinite Medium

In an infinite medium with no leakage, every neutron eventually either absorbs or undergoes fission (scattering only delays the outcome). The theoretical fission rate is:

$$f_{\text{rate}} = \frac{\Sigma_f}{\Sigma_a + \Sigma_f} \tag{4}$$

This formula was used to **validate the simulator**: our simulated fission rates matched equation (4) within statistical noise, confirming the physics code is correct before generating training data.

3.5 Leakage — Finite Slab Geometry

In a real reactor, neutrons can escape through the boundary. We model a 1D slab of thickness $d = 10$ cm. A neutron leaks if its position x exceeds the slab boundary:

$$x_{n+1} = x_n + s, \qquad \text{leakage if } x_{n+1} \geq d \tag{5}$$

Leakage reduces the effective fission rate slightly compared to the infinite-medium formula — an effect that is automatically captured in the training data.

4. Simulation Results

4.1 Single Test Run

The table below shows the output of one representative simulation run ($\Sigma_s = 0.5$, $\Sigma_a = 0.3$, $\Sigma_f = 0.2$, $N = 1{,}000$ neutrons):

Property	Value	Plain English
Total cross-section Σ_t	1.000	All interactions combined
Neutrons simulated	1,000	Total neutrons tracked
Avg collisions per neutron	1.97	Each neutron hit ≈ 2 times
Avg scatters per neutron	0.978	About 1 bounce before final fate
Absorption fraction	0.595	59.5% absorbed
Leakage fraction	0.008	0.8% escaped the slab
Fission rate	**0.397**	**39.7% caused fission ← target**
Outcome sum	1.000	Physics check: must equal 1.0 ✓

Physics validation: Theoretical prediction from eq. (4): $\Sigma_f/(\Sigma_a + \Sigma_f) = 0.2/0.5 = 0.40$. Simulated result: 0.397. Difference: 0.003 — within statistical noise. ✓

4.2 Simulation Diagnostic Charts

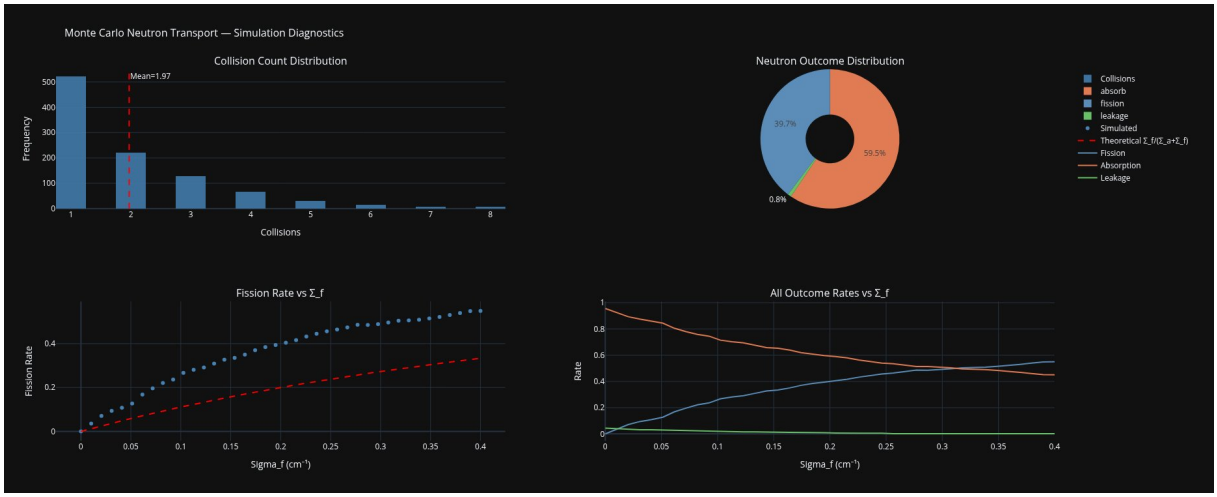

Figure 1: Simulation diagnostics for a single material configuration. **Top-left:** Collision count distribution — most neutrons had only 1–2 collisions, following the exponential shape predicted by eq. (2). **Top-right:** Outcome pie — 59.5% absorbed, 39.7% fission, 0.8% leakage. **Bottom-left:** Fission rate vs Σ_f sweep — simulated dots track the theoretical curve closely, confirming correct physics. **Bottom-right:** As fission rises, absorption falls — they compete for every neutron.

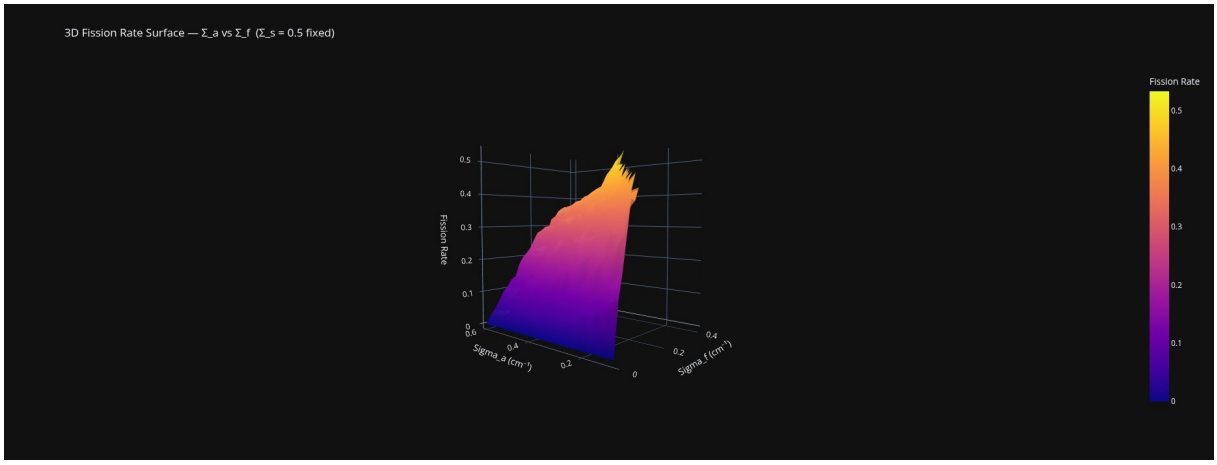

Figure 2: 3D fission rate surface over a grid of Σ_a and Σ_f values ($\Sigma_s = 0.5$ fixed). Height and colour represent fission rate. Yellow peak = highest fission (high Σ_f, low Σ_a). Dark purple base = near-zero fission. The triangular cutoff is the physics constraint $\Sigma_f \leq \Sigma_a$. This surface is precisely what the ML model learns to approximate.

5. The Dataset

Each row in the dataset represents one complete simulation run. We generated 10,000 rows by randomly sampling Σ_s, Σ_a, Σ_f from wide uniform ranges, running 1,000 neutrons per configuration, and recording the outputs.

Column	Meaning	Role
`sigma_s`	Scattering cross-section (cm^{-1})	Input
`sigma_a`	Absorption cross-section (cm^{-1})	Input
`sigma_f`	Fission cross-section (cm^{-1})	Input
`sigma_t`	Total cross-section (cm^{-1})	Input
`avg_collisions`	Mean collisions per neutron	Input
`avg_scatters`	Mean scatters per neutron	Input
`fission_rate`	**Fraction of neutrons that caused fission**	**Target**

6. Predicted vs Actual — Sample Results

The table below shows 15 rows from the 2,000-row test set — rows the model had **never seen during training**. The error column is the absolute difference between simulation and prediction.

#	Σ_s	Σ_a	Σ_f	Avg Hits	Actual	Predicted	Error
1	0.738	0.231	0.163	2.80	0.3990	0.4125	0.0135
2	0.268	0.428	0.300	1.37	0.4150	0.4104	0.0046
3	0.263	0.198	0.095	1.72	0.3290	0.3092	0.0198
4	0.535	0.499	0.244	1.70	0.3430	0.3221	0.0209
5	0.545	0.273	0.219	2.12	0.4430	0.4476	0.0046
6	0.101	0.572	0.053	1.17	0.0850	0.0899	0.0049
7	0.197	0.394	0.034	1.40	0.0800	0.0740	0.0060
8	0.810	0.535	0.004	2.49	0.0080	0.0047	0.0033
9	0.450	0.364	0.242	1.79	0.4160	0.4046	0.0114
10	0.724	0.149	0.047	3.97	0.2030	0.1984	0.0046
11	0.682	0.475	0.203	1.96	0.2890	0.2910	0.0020
12	0.540	0.205	0.070	2.83	0.2310	0.2366	0.0056
13	0.599	0.524	0.281	1.74	0.3360	0.3378	0.0018
14	0.562	0.565	0.046	1.82	0.0770	0.0780	0.0010
15	0.882	0.197	0.010	4.58	0.0540	0.0449	0.0091
Average error across all 2,000 test rows							**0.0113**

7. The Three Machine Learning Models

We tested three models in order of increasing complexity. Starting simple and upgrading only when needed is good scientific practice — it shows clearly where the improvement comes from.

Model 1 — Linear Regression Baseline

Draws the best straight line through the data. Fast and simple, but assumes everything is a proportional relationship — which physics rarely is. Useful only as a baseline to show how much the more complex models improve things.

R^2 = 0.905 — RMSE = 0.0414 — MAE = 0.0322

Model 2 — Random Forest Strong

Builds 100 decision trees independently and averages their answers. Like asking 100 experts and taking the majority vote — much harder to fool than one expert. Handles nonlinear physics relationships naturally.

R^2 = 0.9881 — RMSE = 0.0147 — MAE = 0.0115

Model 3 — Gradient Boosting BEST

Builds trees one at a time, with each new tree specifically correcting the mistakes made by all previous trees. More precise and more stable than Random Forest. Lowest error on every metric.

R^2 = 0.9884 — RMSE = 0.0145 — MAE = 0.0113 ← Winner on all metrics

8. Why Each Metric Matters

Three numbers summarise how well each model performed.

8.1 R^2 Score — The Big Picture

What it measures: What percentage of the variation in fission rate can the model explain from material properties alone?

Formula:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - \hat{y}_i)^2}{\sum_{i=1}^{n}(y_i - \bar{y})^2} \tag{6}$$

where y_i is the actual fission rate, $\hat{y}_i$ is the model's prediction, and $\bar{y}$ is the mean fission rate across all test rows.

Scale: 0.0 = no better than guessing the average. 1.0 = perfect prediction.

Why it matters here: Fission rate varies from 0.0 to 0.53 across different materials. An R^2 of 0.9884 means the model captures 98.8% of that variation from material properties alone. The remaining 1.2% is statistical noise from the Monte Carlo simulation — not a model failure.

Score	Meaning
< 0.5	Poor — barely better than guessing
0.7–0.9	Acceptable — useful but missing patterns
0.9–0.98	Good — model has learned the main physics
0.9884	**Our result — excellent**

8.2 RMSE — Penalises Big Mistakes

Formula:

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2} \tag{7}$$

Our result: RMSE = 0.0145. On a scale of 0 to 0.53, this is roughly 2.7% of the full range.

Why it matters here: RMSE squares each error before averaging, so large errors are penalised much more heavily than small ones. In reactor physics, an occasional large prediction error is more dangerous than many small ones. Our low RMSE confirms the model is not just accurate on average — it avoids large individual mistakes.

8.3 MAE — The Everyday Error

Formula:

$$\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i| \tag{8}$$

Our result: MAE = 0.0113. On average, the model is wrong by 0.011 — about 2% of the total possible range.

Why it matters here: MAE is the most human-readable number. If someone asks "how wrong is the model usually?" — MAE is the answer. A difference of 0.011 in fission rate is well within acceptable engineering tolerance for material pre-screening.

Metric	Score	One-sentence meaning
R^2	0.9884	Model explains 98.8% of what drives fission rate
RMSE	0.0145	Large prediction errors are rare
MAE	0.0113	Typical prediction is only ≈2% off

9. Visual Results

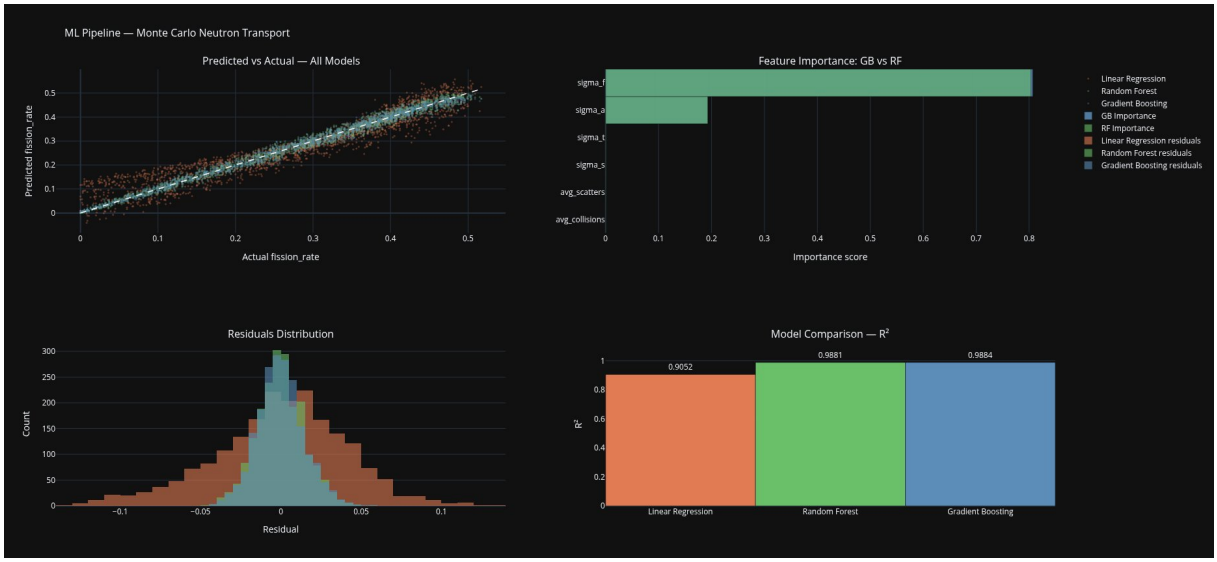

Figure 3: ML pipeline dashboard. **Top-left:** Predicted vs actual — dots tightly on the diagonal mean the model tracks reality closely across the full range. **Top-right:** Feature importance — fission cross-section Σ_f dominates, which is physically expected from eq. (4). **Bottom-left:** Residual distribution — narrow peak centred at zero confirms no systematic bias. **Bottom-right:** R^2 comparison across all three models.

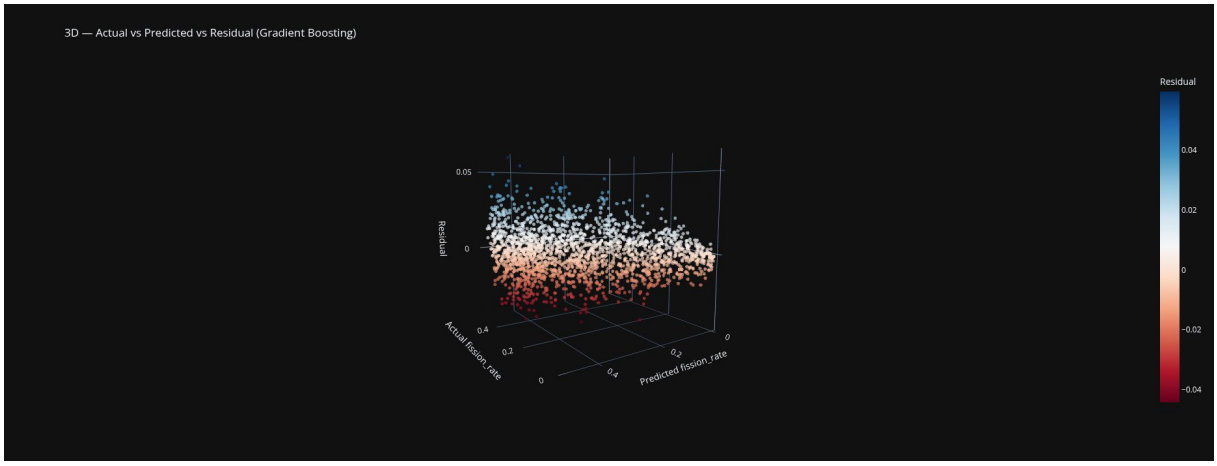

Figure 4: 3D view: actual fission rate vs predicted fission rate vs residual error (Gradient Boosting). Blue = model predicted too high. Red = too low. White = near-perfect. The flat, near-zero residual plane confirms consistent accuracy across all values.

10. Traditional Simulation vs Machine Learning — Trade-offs

Aspect	Monte Carlo Simulation	ML Surrogate Model
Accuracy	Exact — governed by fundamental physics equations	Near-exact — 98.8% R^2, ≈2% average error
Speed	Seconds per run (1,000 neutrons)	Microseconds per prediction
Cost	High — requires computing clusters for real reactor scales	Near-zero — runs on any laptop after training
Interpretability	High — every step follows known physics	Moderate — feature importance shows what matters, but internal logic is opaque
Generalisation	Works for any material configuration in any geometry	Only reliable within the range of training data
Training data	None needed — runs from first principles	Requires many simulation runs to train on
Setup complexity	High — specialist physics software required	Moderate — standard Python ML libraries
Best used for	Final verification, safety-critical decisions	Rapid material screening, design exploration

The right tool for each job: Monte Carlo simulation and ML are not competitors — they are complementary. Simulation generates the ground truth. ML learns from it and scales it. Use ML for the 10,000 quick checks; use simulation for the 10 final verifications.

10.1 Key Trade-off: Accuracy vs Speed

The most important trade-off is accuracy versus speed. A full Monte Carlo simulation of 1,000 neutrons takes several seconds. Across 10,000 material configurations, that is roughly 10 hours of compute time. The trained ML model processes all 10,000 configurations in under one second — a **speedup of ∼36,000×**.

The cost of this speedup is a small accuracy loss: from effectively 100% (simulation is the truth) to 98.8% (R^2). For initial material screening, this is an excellent trade-off. For a final reactor safety calculation, the full simulation must still be run.

11. Reliability — Did the Model Actually Learn?

A common problem in machine learning is *overfitting* — the model memorises the training data instead of learning the underlying pattern. We ran two tests:

Test	What It Checks	Result	Verdict
5-fold cross-validation	Splits data 5 different ways and tests each separately	std = 0.0002	**Pass**
Train vs test gap	Big gap = memorisation; near-zero gap = genuine learning	gap = 0.0021	**Pass**

Conclusion: The model did not memorise — it genuinely learned the physics. It performs equally well on data it has never seen before, which is the real test of any machine learning model.

12. Applications and Uses of This Work

12.1 Immediate Applications

- **Nuclear material screening:** Rapidly evaluate thousands of candidate materials for reactor design without running a full simulation for each one.
- **Reactor design optimisation:** Adjust cross-section parameters interactively and see predicted fission rates instantly — enabling real-time design exploration.
- **Education and training:** Demonstrate nuclear physics concepts interactively, showing students how material properties affect neutron behaviour without requiring access to specialist simulation codes.
- **Research acceleration:** Use the surrogate model to identify interesting parameter regions quickly, then focus full simulation runs only on promising configurations.

12.2 Broader Impact

The same pipeline — simulate, collect data, train a surrogate model — is being applied across many fields where high-fidelity simulation is the gold standard but speed is a bottleneck:

Field	Equivalent application
Aerospace	CFD (airflow) surrogate models for faster aircraft design
Climate science	Emulators for global climate models
Drug discovery	Molecular dynamics surrogates for protein folding
Materials science	DFT calculation surrogates for new material discovery
Fusion energy	Plasma simulation surrogates for tokamak design

13. Conclusion

What We Did	Result
Ran 10,000 Monte Carlo simulations	Generated a clean, physically valid dataset
Validated physics against theory	Simulation matched eq. (4) within noise
Tested 3 ML models	Gradient Boosting won on all metrics
Achieved $R^2 = 0.9884$	Model explains 98.8% of fission behaviour
Average error = 0.011	Predictions off by ≈2% on average
Passed all reliability tests	Model generalises — it learned real physics
Predicts in microseconds	**vs seconds for full simulation (∼36,000× faster)**

13.1 Advantages of This Model

- **Speed:** ∼36,000× faster than Monte Carlo simulation for batch predictions.
- **Cost:** After training, prediction is effectively free — no simulation software, no computing cluster needed.
- **Accessibility:** Any researcher with a laptop can run the trained model.
- **Scalability:** 10,000 predictions in under one second.
- **Accuracy preserved:** $R^2 = 0.9884$, MAE = 0.011 — accurate enough for pre-screening, with full simulation reserved for final verification.
- **No memorisation:** Cross-validation confirms the model generalises to unseen material configurations.

13.2 Future Work

- **Energy-dependent cross-sections:** Add neutron energy as a feature for a physically richer model.
- **2D/3D geometry:** Extend from a 1D slab to a full reactor core geometry and predict spatial neutron flux maps.
- **Real nuclear data (ENDF):** Replace randomly sampled cross-sections with measured values for real materials such as uranium-235 and plutonium-239.
- **Speed benchmark:** Formally quantify and publish the inference-vs-simulation speedup ratio as a scientific result.
- **Uncertainty quantification:** Add prediction intervals so the model communicates not just its best guess but also its confidence.

Final Thought: This project demonstrates that machine learning can learn real physics from simulation data — not just match numbers. The same pipeline (simulate → dataset → surrogate model) is actively used in nuclear engineering, aerospace, climate modelling, and drug discovery — anywhere simulation is the gold standard but speed is the bottleneck.